$A = B$,
or
Everything You Always Wanted To Know About Formulas*
(* But Were Afraid To Ask)

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Disclaimer / introduction

In this talk, I will tell you about various formulas: how to prove them, how to discover them, how to investigate them.
For example,

- $\sum_{k\geq 0} \binom{n}{k} = 2^n$ (we all know that one)
- $\sum_{k\geq 0} \binom{2k}{k}\binom{2n-2k}{n-k} = 4^n$ (can be proved by a little dirty trick)
- already very tricky Dixon's formula (1894)
  $$\sum_{k\geq 0}^{n} (-1)^k \binom{n}{k}^3 = \begin{cases} (-1)^m \frac{(3m)!}{(m!)^3}, & \text{if } n = 2m, \\ 0, & \text{if } n = 2m+1. \end{cases}$$
- $\sum_{k\geq 0} \binom{n-k}{k} = F_{n+1}$ (unusual guise of Fibonacci numbers)

But some formulas will be way outside our scope, so don't sue me if your formula does not fit.

Little dirty trick

Chu (1303) –Vandermonde (1772) formula:

$$\binom{a+b}{n} = \sum_{i+j=n} \binom{a}{i}\binom{b}{j}.$$

Let us try $a = b = -\frac{1}{2}$:

$$\binom{-\frac{1}{2}}{i} = \frac{-\frac{1}{2}(-\frac{1}{2}-1)\cdots(-\frac{1}{2}-i+1)}{i!} = \left(-\frac{1}{4}\right)^i \binom{2i}{i},$$

$$\binom{-1}{i} = \frac{-1(-1-1)\cdots(-1-i+1)}{i!} = (-1)^i,$$

so the formula becomes

$$(-1)^n = \sum_{i+j=n} = \left(-\frac{1}{4}\right)^i \binom{2i}{i} \left(-\frac{1}{4}\right)^j \binom{2j}{j},$$

and that is easily transformed into

$$\sum_{k\geq 0} \binom{2k}{k}\binom{2n-2k}{n-k} = 4^n.$$

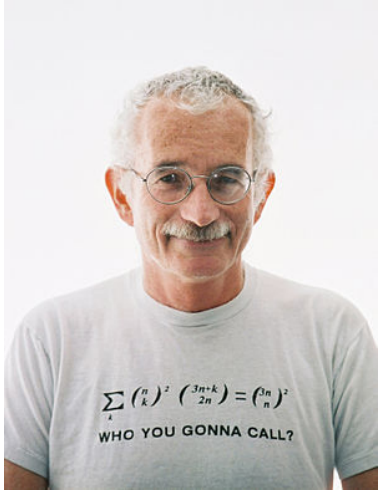
A trip down the memory lane

The number of $n \times n$-matrices X of integers modulo p satisfying $X^2 = 0$ (p a prime number) is equal to a certain remarkable polynomial $C_n(p)$.

But the margin of this slide is too small for the elegant formula...

This was conjectured by A. A. Kirillov (1995), and proved by D. Zeilberger and [the Zeilberger's computer] Shalosh B. Ekhad (1996).

Alexandre Kirillov (b. 1936)



Doron Zeilberger (b. 1950)

A trip down the memory lane

Theorem. The number of $n \times n$-matrices X of integers modulo p satisfying $X^2 = 0$ is given by the formula

$$C_n(p) = \sum_{\substack{s \equiv n+1 \pmod 2 \\ s \equiv (-1)^n \pmod 3}} \left(\binom{n}{\frac{n+1-s}{2}} - \binom{n}{\frac{n-1-s}{2}} \right) p^{\frac{n^2}{4} + \frac{1-s^2}{12}}$$

A trip down the memory lane

Proof. Denote by $C_n^r(p)$ the number of such matrices of size r. By relatively simple linear algebra,

$$C_{n+1}^{r+1}(p) = p^{r+1} C_n^{r+1}(p) + (p^{n-r} - p^r) C_n^r(p),$$

and $C_n(p) = \sum_{r \geq 0} C_n^r(p)$. Then one notes that C_n^r is equal to the constant term in z of

$$\frac{(1-z)(1+z)^n p^{r(n-r)}}{z^r} \sum_{i \geq 0} (-1)^i p^{-\binom{i+1}{2} - i(n-2r)} \binom{i+n-2r}{i}_p z^i$$

(here $\binom{n}{k}_p = \frac{(p-1)(p^2-1)\cdots(p^n-1)}{(p-1)(p^2-1)\cdots(p^k-1)(p-1)(p^2-1)\cdots(p^{n-k}-1)}$).
Finally, we compute the sum, change the order of summation, and apply the general "$A = B$" method of Wilf and Zeilberger.

Another example: the zeta function

Zeta function:

$$\zeta(s) = \sum_{n \geq 1} \frac{1}{n^s}.$$

Euler computed $\zeta(2k)$ for all positive integer k, e.g. $\zeta(2) = \frac{\pi^2}{6}$. Riemann proved that $\zeta(s)$ can be analytically extended to all *complex* $s \neq 1$, and conjectured that apart from the 'trivial' solutions $s = -2k$, all solutions to $\zeta(s) = 0$ must have $\mathrm{Re}(s) = \frac{1}{2}$. That statement, the *Riemann Hypothesis*, is an important unsolved problem in mathematics.

Leonhard Euler (1707-1783)

Bernhard Riemann (1826–1866)

Values of $\zeta(n)$

Euler proved that $\zeta(2k)$ is a rational multiple of π^{2k}. The values $\zeta(2k+1)$ remained completely mysterious until relatively recently.

In 1977, a French mathematician R. Apéry proved irrationality of $\zeta(3)$. His strategy was to produce some remarkable formulas for rational approximations to $\zeta(3)$ which were "too good to be true" in the case it were rational.

Roger Apéry (1916–1994)

Apéry's proof

Consider a finite difference equation

$$n^3 x_n + (n-1)^3 x_{n-2} = (34n^3 - 51n^2 + 27n - 5)x_{n-1}.$$

Define $\{A_n\}$ as the solution to this equation with $A_0 = 1$ and $A_1 = 5$, then all A_n are integers (!!). For instance $A_2 = 73$, $A_3 = 1445$, $A_4 = 33001$. Define $\{B_n\}$ as the solution to this equation with $B_0 = 0$, $B_1 = 6$, then B_n is a rational number with the denominator bounded by the $2 \cdot \mathrm{lcm}(1, 2, \ldots, n)^3$.

Moreover, we have

$$\lim_{n\to\infty} \frac{A_n}{B_n} = \zeta(3),$$

and

$$0 < \left|\zeta(3) - \frac{A_n}{B_n}\right| < \frac{c}{B_n^2}$$

for some $c > 0$. This readily implies that $\zeta(3)$ is irrational.

Apéry's proof

In fact, Apéry proved more. He established that the numbers

$$A_n = \sum_k \binom{n}{k}^2 \binom{n+k}{k}^2$$

verify $A_0 = 1$, $A_1 = 5$, and

$$n^3 A_n + (n-1)^3 A_{n-2} = (34n^3 - 51n^2 + 27n - 5)A_{n-1}.$$

He also proved a similar result related to $\zeta(2)$: if

$$a_n = \sum_k \binom{n}{k}^2 \binom{n+k}{k}.$$

Then

$$n^2 a_n - (n-1)^2 a_{n-2} = (11n^2 - 11n + 3)a_{n-1}.$$

Apéry's proof

Two other formulas that Apéry presented in his talk in Luminy in 1977, shocking the audience by all the *deus ex machina* tools he used were

$$3\sum_{n\geq 1}\frac{1}{n^2\binom{2n}{n}}=\sum_{n\geq 1}\frac{1}{n^2}$$

$$\frac{5}{2}\sum_{n\geq 1}\frac{(-1)^{n-1}}{n^3\binom{2n}{n}}=\sum_{n\geq 1}\frac{1}{n^3}$$

How does one obtain all those formulas?

There were several people whose contribution to dealing with formulas like that completely changed the game.

1946: Sister Mary Celine Fasenmyer defended her PhD thesis "Some Generalized Hypergeometric Polynomials", where she introduced a method of finding recurrent relations for "hypergeometric polynomials".

Sister Mary Celine Fasenmyer (1906–1996)

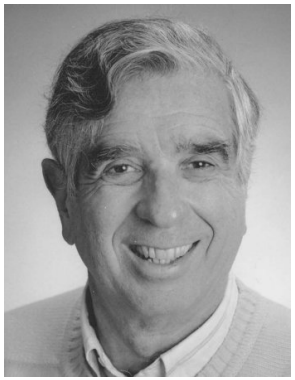
How does one obtain formulas like that?

1977: Bill Gosper published a paper "Decision procedure for indefinite hypergeometric summation" (Proceedings NAS **75** (1978), no. 1, 40–42), where he established an algorithm, finding, for a given sequence $\{a_n\}$, a representation $a_n = S_n - S_{n-1}$, where $\frac{S_n}{S_{n-1}}$ is a rational function of n.

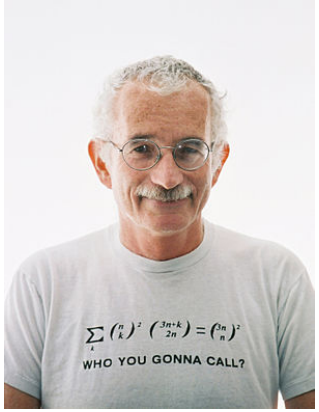
William Gosper (b. 1943)

How does one obtain formulas like that?

In 1980s, Doron Zeilberger published a series of papers improving both Sister Celine's methods and Gosper's methods (e.g. "All binomial identities are verifiable", Proc. NAS **78** (1981), no. 7, p. 4000, and "Sister Celine's technique and its generalisations", J. Math. Analysis Appl. **85** (1982), 114–145)
This attracted attention of Herbert Wilf, who, together with Zeilberger, created what is now called the Wilf–Zeilberger method ("Rational functions certify combinatorial identities", J. Amer. Math. Soc. **3** (1990), no. 1, 147–158, described by Zeilberger on his webpage as "the Steele-prize-winning WZ classic".)



Herbert Wilf (1931–2012)



Doron Zeilberger (b. 1950)

How does one obtain formulas like that?

Essentially, all these papers deal with what is now known as "creative telescoping", which hunts, given a two-index sequence $\{F_{n,k}\}$, for a formula

$$s_0(n)F_{n,k} + \cdots + s_l(n)F_{n+l,k} = G_{n,k} - G_{n,k+1}.$$

For instance, for $F_{n,k} = \binom{n}{k}^2 \binom{n+k}{k}^2$, we can take $l = 2$ and

$$G_{n,k} = 4(2n+1)(k(2k+1) - (2n+1)^2)\binom{n}{k}^2 \binom{n+k}{k}^2,$$

then the $s_i(n)$ come from the LHS of the Apéry's formula.

The small print

As I said in the beginning, only some formulas are OK. What are they?

Hypergeometric terms: we say that $F_{n,k}$ is a proper hypergeometric term if it can be written as

$$F_{n,k} = P(n,k)\frac{\prod_{i=1}^{U}(a_i n + b_i k + c_i)!}{\prod_{j=1}^{V}(u_i n + v_i k + w_i)!}x^k,$$

where $U, V \geq 0$ are integers, a_i, b_i, u_i, v_i are integers not depending on any extra parameters, and P is a polynomial.
For example,

$$\frac{1}{n+3k+1} = \frac{(n+3k)!}{(n+3k+1)!},$$

so $\frac{1}{n+3k+1}$ is a proper hypergeometric term. Meanwhile, $\frac{1}{nk+1}$ is not a proper hypergeometric term.

Sister Celine's method

Theorem. Let $F_{n,k}$ be a proper hypergeometric term. Then it satisfies a k-free recurrence relation: there exist positive integers I and J and polynomials $a_{i,j}(n)$ such that

$$\sum_{i=0}^{I}\sum_{j=0}^{J} a_{i,j}(n)F_{n-i,k-j} = 0.$$

Moreover, we can take

$$J = \sum_s |b_s| + \sum_s |v_s|, I = 1 + \deg(P) + J(-1 + \sum_s |a_s| + \sum_s |u_s|).$$

Sister Celine's method

For example, for $F_{n,k} = \binom{n}{k}$, we get the expected

$$\binom{n}{k} - \binom{n-1}{k} - \binom{n-1}{k-1} = 0.$$

For $F_{n,k} = \binom{n}{k}^2$, the result is more tricky:

$$n\binom{n}{k}^2 - (2n-1)\left(\binom{n-1}{k}^2 + \binom{n-1}{k-1}^2\right) + \\ (n-1)\left(\binom{n-2}{k}^2 - 2\binom{n-2}{k-1}^2 + \binom{n-2}{k-2}^2\right) = 0.$$

Sister Celine's method

Summing

$$n\binom{n}{k}^2 - (2n-1)\left(\binom{n-1}{k}^2 + \binom{n-1}{k-1}^2\right) + \\ (n-1)\left(\binom{n-2}{k}^2 - 2\binom{n-2}{k-1}^2 + \binom{n-2}{k-2}^2\right) = 0.$$

over all k, we get, for $d_n = \sum_{k\geq 0} \binom{n}{k}^2$, the recurrence relation

$$nd_n - (2n-1)(d_{n-1} + d_{n-1}) + (n-1)(d_{n-2} - 2d_{n-2} + d_{n-2}) = 0,$$

which is just $d_n = \frac{2(2n-1)}{n} d_{n-1}$, easily implying $d_n = \binom{2n}{n}$.

Sister Celine's method

Similarly, for

$$g_n = \sum_{k \geq 0} \binom{n}{k}^3$$

we obtain the recurrence relation

$$n^2 g_n - 8(n-1)^2 g_{n-2} = (7n^2 - 7n + 2) g_{n-1},$$

which rules out hope for a simple formula.

A very recent example (March 2016)

In my own recent work, I encountered the following sequence of numbers:

$$E_n = \frac{1}{7n+1} \sum_{k=0}^{\lfloor n/3 \rfloor} (-1)^k \binom{7n+k}{k} \binom{7n+1}{n-3k}.$$

Applying the Zeilberger's Algorithm, we find that

$$s_0(n)E_n - s_1(n)E_{n-1} + s_2(n)E(n-2) = 0,$$

where s_i are certain polynomials of degree 20 with coefficients of magnitude about 10^{18}. A rather subtle analysis of this recurrence relation can lead to a proof of the fact that E_n are actually positive numbers, which is not at all clear from their definition.
The argument is similar to the method of Apéry, but the number appearing instead of $\zeta(3)$ is a rational number simply equal to 1.

That's all folks!

Thank you for your patience!